

SUPPLEMENTARY INFORMATION

Topology-enhanced machine learning for consonant recognition

1 PHONETIC DATA, AUDITORY PERCEPTION, AND LEARNING TOPOLOGICALLY

In this section, we first review the basics of phonetic data, our main objects of study, and explain our scientific approach towards a distribution space for them based on their topological features (rather than biomechanical production). We then review the mechanism of human auditory perception, especially the structure of a cochlea as a biological Fourier analysis machine. This underpins existing audio and speech signal processing technology. In contrast, our topological approach to phonetic data extends beyond biomimetic engineering to more comprehensive, robust feature extraction and learning, as demonstrated with TopNN from Results in the main text.

1.1 Phonetic data and their distribution

As a research field of linguistics, phonetics studies the production as well as the classification of human speech sounds from the world's languages. In phonetics, a *phoneme* is the smallest basic unit of human speech sounds. It is a short speech segment possessing distinct physical or perceptual properties. In the main text and Supplementary Information, to differentiate the theoretical and real-world objects, we reserve the word *phone* for a phoneme segmented from a recording of human speech.

Phonemes are generally classified into two principal categories: vowels and consonants. A *vowel* is defined as a speech sound pronounced by an open vocal tract with no significant build-up of air pressure at any point above the glottis, and at least making some airflow escape through the mouth. In contrast, a *consonant* is a speech sound that is articulated with a complete or partial closure of the vocal tract and usually forces air through a narrow channel in one's mouth or nose.

Unlike vowels which must be pronounced by vibrated vocal cords, consonants can be further categorised into two classes according to whether the vocal cords vibrate or not during articulation. If the vocal cords vibrate, the consonant is known as a *voiced* consonant. Otherwise, the consonant is *voiceless* (or *unvoiced*). Since vocal cord vibration can produce a stable periodic signal of air pressure, voiced consonants tend to have more periodic components than voiceless consonants, which can in turn be detected by PH as topological characteristics from phonetic time series data.

Indeed, one of the more heuristic motivations for our research project is to re-examine (and even revise) the linguistic classifications of phonemes through the mathematical lens of topological patterns and shape of speech data, analogous to Carlsson and his collaborators' seminal work [1] on the distribution of image data (cf. the distribution chart of vowels created by linguists in [2]).

1.2 Spectral signal processing and beyond

The transmission of sound to the human auditory system is a marvel of biological engineering, where acoustic waves are progressively transformed into neural signals. This process begins with the external ear channelling sound waves to the tympanic membrane, which subsequently induces vibration in the ossicles of the middle ear (the malleus, incus, and stapes, constituting the smallest bones in the human body). These minute oscillations are then conveyed to one of the most critical structures in auditory perception: the cochlea.

The cochlea, in essence, functions as a biological Fourier analysis machine (see [3] for an illustration). This spiral-shaped, fluid-filled organ amplifies the incoming sound waves and performs a spectral decomposition of complex acoustic signals. The cochlea's architecture is characterised by a gradual variation in the radius of its spiral and the corresponding mechanical properties of the basilar membrane that runs along its length. The basal end of the cochlea, with its rigid basilar membrane and narrow duct, is optimally tuned to high-frequency vibrations. In contrast, the apical region, featuring a more flexible membrane and wider duct, is more responsive to signals of lower frequency.

This structural gradient creates a tonotopic organisation within the cochlea, analogous to the varying tensions of musical strings producing different pitches. The basilar membrane's varying mechanical properties result in different regions having distinct resonant frequencies, each maximally sensitive to a specific range of sound frequencies. Atop this membrane reside the hair cells, specialised mechanoreceptors that transduce mechanical vibrations into electrical signals and enable auditory perception.

The cochlea's spiral configuration, in conjunction with the basilar membrane's properties, constitutes a natural, passive mechanical Fourier analyser. This biological mechanism effectively distributes frequency components of sound waves along the length of the cochlea. Consequently, the neural signals generated by hair cells at different locations along the basilar membrane correspond to distinct frequency bands of the original acoustic input.

It is noteworthy that contemporary industrial approaches to speech signal processing, such as STFT and MFCC as compared in Results of the main text, employ analytical methods that parallel the cochlea's function. These techniques decompose signals into linear combinations of basis functions, mirroring the cochlea's spectral analysis. This convergence of biological design and signal processing methodology can be viewed as a triumph of biomimetic engineering.

Intriguingly, our experimental findings have demonstrated that topological principles can also be leveraged to extract certain acoustic information. This approach lacks a direct physiological counterpart in current auditory research and established theoretical frameworks. The potential for topological methods in auditory signal processing opens up an exciting frontier for exploration, bridging the gap between abstract mathematics and biological sensory systems. Future investigation in this domain may yield insights that could transform our understanding of auditory perception and inspire innovative signal processing techniques (cf. [4, 5, 6]).

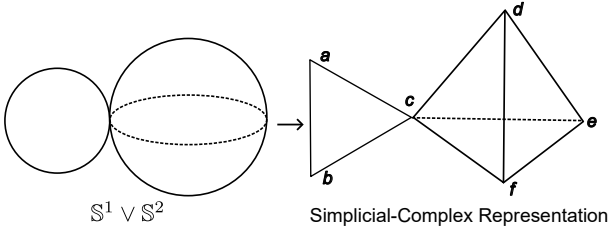
2 PERSISTENT HOMOLOGY

Topology is a subject that studies the properties of geometric objects that remain unchanged under continuous transformations or smooth perturbations. It focuses on the intrinsic features of a space regardless of its rigid shape or size. Algebraic topology provides a quantitative description of these topological properties.

2.1 Simplicial complexes

The structure of a *simplicial complex* (and its numerous variants and analogues) provides a powerful tool in algebraic topology which enables us to represent a topological space using discrete data. Unlike the original space, which can be challenging to compute and analyse, a simplicial complex provides a combinatorial description that is much more amenable to computation. We can use algebraic techniques to study the properties of a simplicial complex, such as its homology and cohomology groups, which encode and reveal information about the topology of the represented space.

Formally, a simplicial complex with *vertices* in a set V is a collection K of nonempty finite subsets $\sigma \subset V$ such that any nonempty subset τ of σ is also contained in K (called a *face* of σ) and that $\sigma, \sigma' \in K$ implies their intersection is either empty or a face of both. A set $\sigma \in K$ with $(i + 1)$ elements is called an *i -simplex* of the simplicial complex K .



Supplementary Figure 1. **From a topological space to its triangulation.**

For instance, Supplementary Fig. 1 illustrates $S^1 \vee S^2$, a circle kissing a sphere at a single point. This topological space can be represented by a simplicial complex K with six vertices a, b, c, d, e , and f , namely,

$$K = \{ \{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \{f\}, \\ \{a, b\}, \{a, c\}, \{b, c\}, \{c, d\}, \{c, f\}, \{d, f\}, \{c, e\}, \\ \{d, e\}, \{f, e\}, \\ \{c, d, f\}, \{c, e, f\}, \{c, d, e\}, \{d, e, f\} \}$$

It is a combinatorial avatar for $S^1 \vee S^2$ via a triangulation operation on the latter, which is amenable to algebraic computation as follows.

2.2 Homology groups and Betti numbers

Let p be a prime number and $\mathbb{F}_p$ be the field with p elements. Given a simplicial complex K , define $C_i(K; \mathbb{F}_p)$ to be the $\mathbb{F}_p$ -vector space with basis the set of i -simplices in K . To keep track of the order of vertices within a simplex, we use the alternative notation with square brackets in the following.

If $\sigma = [v_0, v_1, \dots, v_i]$ is an i -simplex, define the *boundary* of σ , denoted by $\partial\sigma$, to be the alternating sum of the $(i - 1)$ -dimensional faces of σ given by

$$\partial\sigma := \sum_{k=0}^i (-1)^k [v_0, \dots, \widehat{v_k}, \dots, v_i]$$

where $[v_0, \dots, \widehat{v_k}, \dots, v_i]$ is the k 'th $(i - 1)$ -dimensional face of σ missing the vertex v_k . We can extend ∂ over $C_i(K; \mathbb{F}_p)$ as an $\mathbb{F}_p$ -linear operator so that $\partial: C_i(K; \mathbb{F}_p) \rightarrow C_{i-1}(K; \mathbb{F}_p)$. The composition of boundary operators satisfies $\partial \circ \partial = 0$. The elements in $C_i(K; \mathbb{F}_p)$ with boundary 0 are called *i -cycles*. They form a subspace of $C_i(K; \mathbb{F}_p)$, denoted by $Z_i(K; \mathbb{F}_p)$. The elements in $C_i(K; \mathbb{F}_p)$ that are the images under ∂ of elements in $C_{i+1}(K; \mathbb{F}_p)$ are called *i -boundaries*. They form a subspace too, denoted by $B_i(K; \mathbb{F}_p)$. It follows from $\partial \circ \partial = 0$ that

$$B_i(K; \mathbb{F}_p) \subset Z_i(K; \mathbb{F}_p)$$

Then define the quotient space

$$H_i(K; \mathbb{F}_p) := Z_i(K; \mathbb{F}_p) / B_i(K; \mathbb{F}_p)$$

to be the *i 'th homology group of K with $\mathbb{F}_p$ -coefficients*. We call $\dim(H_i(K; \mathbb{F}_p))$ the *i 'th Betti number*, denoted by $\beta_i(K)$, which counts the number of i -dimensional loops (bounding an $(i - 1)$ -dimensional void) in the corresponding topological space. As such, these homology groups are also called the homology groups of the space (it can be shown that they are independent of the particular ways in which the space is triangulated). For example, the Betti numbers of $S^1 \vee S^2$ above are $\beta_1 = 1$, $\beta_2 = 1$, and $\beta_i = 0$ when $i \geq 3$.

2.3 Persistent homology and its representations

The usefulness of these invariants, besides their computability (essentially Gaussian elimination in linear algebra), lies in their tractability along deformations. Given two simplicial complexes K and L , a simplicial map $f: K \rightarrow L$ (that preserves the simplicial structure) induces an $\mathbb{F}_p$ -linear map $H_i(f; \mathbb{F}_p): H_i(K; \mathbb{F}_p) \rightarrow H_i(L; \mathbb{F}_p)$. Thus, if two spaces are topologically equivalent (in fact, being homotopy equivalent suffices), their homology groups must be isomorphic and the Betti numbers match up.

Let (X, d) be a finite point cloud with metric d . Define a family of simplicial complexes, called *Rips complexes*, by

$$R_\epsilon(X) := \{ \sigma \subset X \mid d(x, x') \leq \epsilon \text{ for all } x, x' \in \sigma \}$$

where $\epsilon \geq 0$ is a parameter. The family

$$\mathcal{R}(X) := \{ R_\epsilon(X) \}_{\epsilon \geq 0}$$

is known as the *Rips filtration* of X . Clearly, if $\epsilon_1 \leq \epsilon_2$, then $R_{\epsilon_1}(X) \hookrightarrow R_{\epsilon_2}(X)$. Thus, for each i we obtain a sequence

$$H_i(R_{\epsilon_0}(X); \mathbb{F}_p) \rightarrow H_i(R_{\epsilon_1}(X); \mathbb{F}_p) \rightarrow \dots \\ \rightarrow H_i(R_{\epsilon_m}(X); \mathbb{F}_p)$$

where $0 = \epsilon_0 < \epsilon_1 < \dots < \epsilon_m < \infty$. As ϵ varies, the topological features of the simplicial complexes $R_\epsilon(X)$ vary, resulting in the emergence and disappearance of loops (see Fig. 1d).

Given the values of ϵ , record the instances of emergence and disappearance of loops, which correspond to cycle

classes in the homology groups along the above sequence. Thus each class has a descriptor $(b, d) \in \mathbb{R}^2$, where b represents its *birth time*, d represents its *death time*, and $d - b$ represents its *lifetime*. In this way, we obtain a multiset

$$\{(b_j, d_j)\}_{j \in J} =: \text{dgm}_i(\mathcal{R}(X))$$

which encodes the persistence of topological features of X . This multiset can be represented as a multiset of points in the 2-dimensional coordinate system, called a *persistence diagram* for the i 'th PH, or as an array of interval segments, called a *persistence barcode* (see Fig. 1f). In particular, by *maximal persistence* we refer to the maximal lifetime among all the points in a persistence diagram. For convenience, sometimes we adopt a variant of the persistence diagram whose vertical axis represents lifetime instead of death time, such as in Fig. 6a.

3 TIME-DELAY EMBEDDING

Time-delay embedding (TDE) is also known as Takens's embedding, sliding window embedding, delay embedding, and delay coordinate embedding. For simplicity, we focus on 1-dimensional time series.

3.1 Continuous time series

TDE of a real-valued function $f: \mathbb{R} \rightarrow \mathbb{R}$, with parameters positive integer d and positive real number τ , is defined to be the vector-valued function

$$\mathcal{E}_{d,\tau}f: \mathbb{R} \rightarrow \mathbb{R}^d \\ t \mapsto (f(t), f(t+\tau), \dots, f(t+(d-1)\tau))$$

Here, d is the *dimension* of the target space for the embedding, τ is the *delay*, and their product $\tau \cdot d$ is called the *window size*.

Let M be a compact smooth manifold of dimension m , $\{\phi_t: M \rightarrow M\}_{t>0}$ be a smooth dynamical system, and $G: M \rightarrow \mathbb{R}$ be a smooth (observation) function. According to Takens's embedding theorem [7], for generic pairs (ϕ_t, G) and a fixed $\tau > 0$, the delay-coordinate map $\Psi: M \rightarrow \mathbb{R}^d$ defined by

$$\Psi(x) = (G(x), G(\phi_\tau(x)), G(\phi_{2\tau}(x)), \dots, G(\phi_{(d-1)\tau}(x)))$$

is an embedding, provided that $d \geq 2m + 1$.

Therefore, with the assumption that our time series data is generated by an unknown dynamical system evolving on a smooth manifold and an unknown observation function, i.e., $f(t) = G(\phi_t(x_0))$, the image $\mathcal{E}_{d,\tau}f$ of the TDE reconstructs the topological shape of the trajectory of the base point x_0 in the manifold M up to homeomorphism, provided the condition $d \geq 2m + 1$. In particular, when the trajectory converges to an attractor, the reconstruction quality improves significantly. This is because attractors are invariant sets, in the sense that once a trajectory enters an attractor, it remains within it, and nearby trajectories asymptotically approach it. Moreover, attractors are minimal, that is, they cannot be decomposed into smaller invariant subsets. Consequently, the reconstructed point cloud becomes denser near the attractor, giving a more faithful representation of the underlying dynamics.

In [8, Sec. 5], Perea and Harer established that the n -truncated Fourier series expansion

$$S_n f(t) = \sum_{k=0}^n a_k \cos(kt) + b_k \sin(kt)$$

of a periodic time series f can be reconstructed into a circle when $d \geq 2n$, i.e.,

$$\mathcal{E}_{d,\tau} S_n f(\mathbb{R}) \cong \mathbb{S}^1$$

Moreover, let L be a constant such that

$$f\left(t + \frac{2\pi}{L}\right) = f(t)$$

Then the 1-dimensional MP of the resulting point cloud is the largest when the window size $\tau \cdot d$ is integrally proportional to $2\pi/L$, i.e.,

$$\tau \cdot d = m \cdot \frac{2\pi}{L}$$

for a positive integer m . Intuitively, an increase in the embedding dimension results in a better approximation when truncating the Fourier series, and the MP of the point cloud becomes the most significant when the window size equals a period.

This methodology also proves particularly advantageous in scenarios where the system under investigation exhibits nonlinear dynamics, precluding straightforward analysis of the time series data. Via a suitable embedding, the inherent geometric configuration of the system emerges, enabling deeper comprehension and refined analysis.

3.2 Discrete time series

In practice, we work with discrete-time signals obtained by sampling a continuous function over a fixed time interval, which we often normalise to be of length $\Delta t = 1$. Suppose that the number of samples equals N , and write $[N]$ for the set $\{0, 1, \dots, N-1\}$. Given a discrete signal $f: [N] \rightarrow \mathbb{R}$, we can apply TDE directly to the sequence, by taking the delay parameter τ to be a positive integer.

It is also common to introduce an additional parameter of *skip* s , with s a positive integer, that controls the step between successive starting points of windows. In effect, with this skip we down-sample the signals by taking every s windows (of size $\tau \cdot d$), reducing the amount of data and computational load. This procedure can also act as a simple low-pass filter that attenuates high-frequency noise.

Given skip s , the set T_s of *admissible window starting points* consists of those sample times t such that a window of d points beginning at t and evenly spaced by τ stays within the signal range $[N]$, and such that these times t are evenly spaced by s . Typically, a sampling begins with $t = 0$, so that

$$T_s = \{t \in [N] \mid t + (d-1)\tau \leq N-1 \text{ and } s \text{ divides } t\}$$

The corresponding TDE is then

$$\mathcal{E}_{d,\tau,s}f: T_s \rightarrow \mathbb{R}^d \\ t \mapsto (f(t), f(t+\tau), \dots, f(t+(d-1)\tau))$$

It converts the signal f into a point cloud in $\mathbb{R}^d$, with one d -dimensional vector for each window starting point $t \in T_s$.

For example, let us consider a signal sequence $f(0), f(1), \dots, f(9)$ with $N = 10$, and choose embedding parameters $d = 3$, $\tau = 2$, and $s = 3$ so that $T_s = \{0, 3\}$. The embedding then yields two points in $\mathbb{R}^3$, namely,

$$\begin{aligned}\mathcal{E}_{3,2,3}f(0) &= (f(0), f(2), f(4)) \text{ and} \\ \mathcal{E}_{3,2,3}f(3) &= (f(3), f(5), f(7))\end{aligned}$$

3.3 Cyclic time-delay embedding

While standard TDE offers a fundamental approach for state space reconstruction, its parameter selection faces critical constraints requiring the sample size (i.e., number of sampled points) $N \geq (d - 1)\tau$. Given a discrete signal $f: [N] \rightarrow \mathbb{R}$, conventional TDE with parameters d and τ generates at most $N - (d - 1)\tau$ embedded points (with skip $s = 1$), which creates a sample-size dilemma that severely restricts practical applications with small series length N or large values of d and τ .

To overcome this limitation, we propose a reconstruction method of *cyclic time-delay embedding* (CTDE). By implementing cyclic boundary conditions through modular arithmetic, CTDE preserves the complete dataset without truncation. Our method enables nearly unrestricted parameter selection, allowing d and τ to independently vary over the full parameter space from 1 to N . Crucially, CTDE maintains a sample size of N embedded points regardless of parameter choices (if the skip $s = 1$), which yields a consistent and unbiased platform for systematic parameter optimisation. Formally, the CTDE mapping is defined as follows:

$$\begin{aligned}\mathcal{CE}_{d,\tau,s}f: T_s &\rightarrow \mathbb{R}^d \\ t &\mapsto \left(f(t \bmod N), f(t + \tau \bmod N), \dots, \right. \\ &\quad \left. f(t + (d - 1)\tau \bmod N) \right)\end{aligned}$$

where $r = a \bmod N$ denotes the standard modulo operation that returns the unique integer $r \in [0, N)$ satisfying $a = kN + r$ for some integer k .

Supplementary Fig. 2 shows embedded point clouds generated by both standard and cyclic TDE methods, displaying their 3D PCA projections across various embedding dimensions and time delays. The sample of a voiced consonant is the same as the one in Fig. 8a.

4 METHODS WITH TOPCAP

In this section, we provide details of some of the methods applied when carrying out the experiments in the main text, particularly TopCap (cf. Fig. 3e).

4.1 Deriving phonetic data from natural speech

We use natural-speech files sourced from the ALLSTAR Corpus of SpeechBox [9], task HT1 language English L1, retrieved on 28th January 2023. SpeechBox is a web-based system providing access to an extensive collection of digital speech corpora developed by the Speech Communication Research Group in the Department of Linguistics at Northwestern University.

The HT1–English section contains a total of 25 individual files, comprising 14 files from women and 11 files from

men. The age range of these speakers spans from 18 to 26 years, with an average of 19.92. Each file is presented in the WAV format and is accompanied by its corresponding aligned file in Textgrid format, which features three tiers of sentences, words, and phones. Collectively, these 25 speech files amount to a total duration of 41.21 minutes. Each speech file contains an individual reading the same sentences consecutively for a duration ranging from 80 to 120 seconds, contingent upon their pace.

The original .wav file has a sampling frequency of 22050 Hz and comprises only one channel. Since MFA [10] is trained in a sampling frequency of 16 kHz, we opt to adjust the sampling frequency of the .wav files accordingly. We then extract the words tier from Textgrid and align words into phones using English_MFA dictionary and acoustic model (MFA version 2.0.6). Thus we obtain corresponding phonetic data from these speech files.

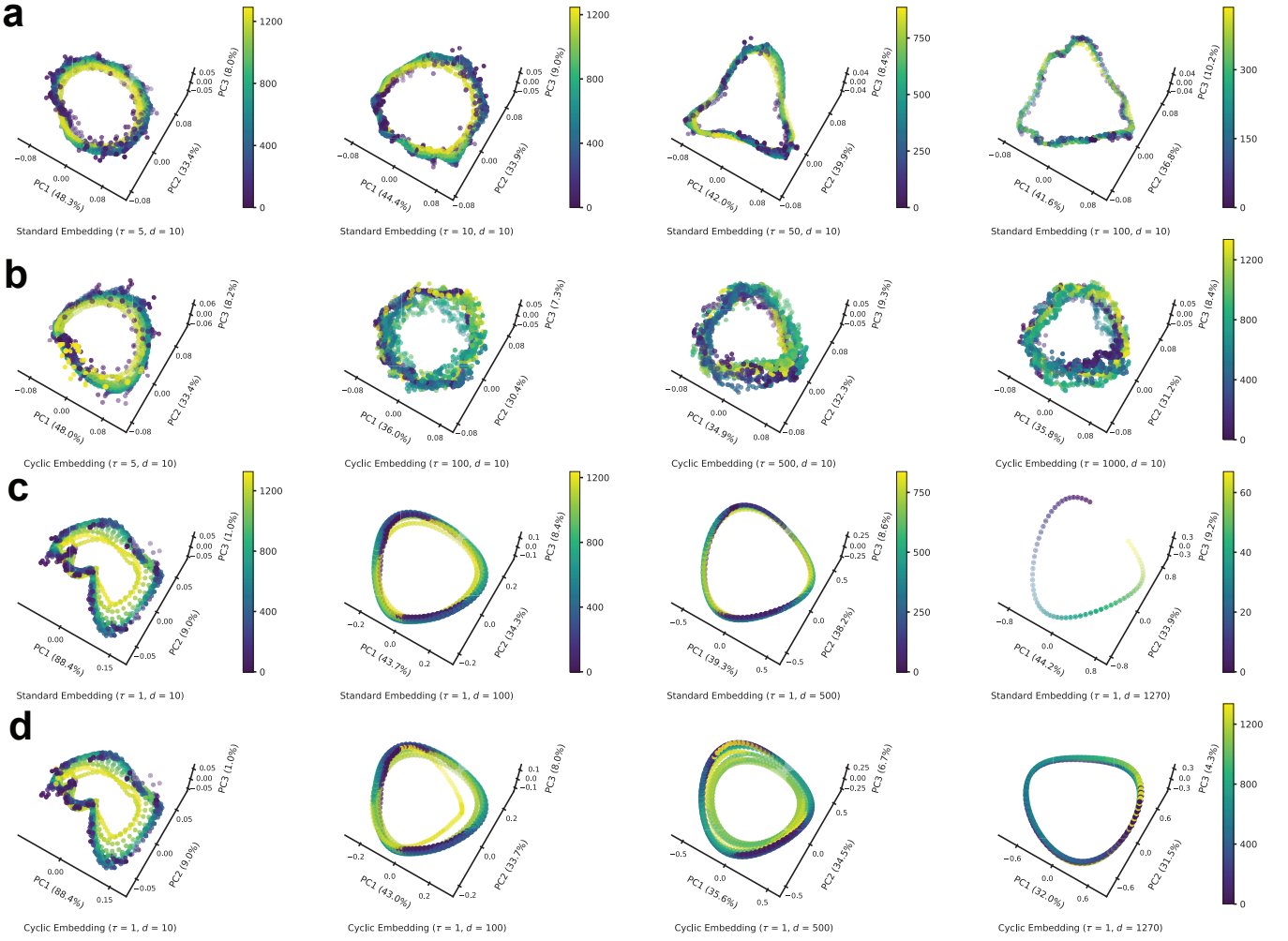
Subsequently, we use voiced and voiceless consonants in those segments as our dataset. Voiced consonants are consonants for which vocal cords vibrate in the throat during articulation, while voiceless consonants are pronounced otherwise (see Supplementary Sec. 1 for more details). Specifically, using Praat [11], we extract voiced consonants [ɪ], [m], [n], [l], [v], and [ʒ]. For voiceless consonants, we select [f], [k], [θ], [t], [s], and [tʃ]. Our selection is limited to these voiced and voiceless consonants, as we aim to balance the ratio of voiced and voiceless consonant records in these speech files. Additionally, some consonants, such as [d] and [h], appear difficult to classify by our methods.

4.2 Selecting parameters for time-delay embedding

Before extracting topological features from a time series, we first endow this 1-dimensional time series with a (Euclidean) topological structure through TDE (see Supplementary Sec. 3). It is noteworthy that this technique also applies to multi-dimensional time series. The ambient space throughout this article is always a Euclidean space. By establishing the topological structure there, or more precisely, the distance matrices, we subsequently compute PH.

For TDE, we select suitable parameters to capture the theoretically optimal MP of a given time series. Here, the embedding dimension is fixed to be 100. Our principle for determining an appropriate dimension is that we want to choose the embedding dimension to be large for a time series of limited length. As discussed in Supplementary Sec. 3 and Discussion in the main text, a higher dimension results in a more accurate approximation by sampled data points. This approach also aims at enhancing computational efficiency and the occurrence of more prominent MP.

Nonetheless, it is important to exercise caution when selecting the dimension, as excessively large dimensions may lead to empty point clouds and other uncontrollable factors. For instance, with a time series consisting of approximately 1200 points, setting the dimension to 100, delay to 5, and skip to 1 results in around 700 points in the corresponding point cloud in Euclidean space. However, increasing the dimension to 200 with the same remaining parameters yields only 200 points, which may be less sufficient to adequately represent the original data structure. Thus, the dimension is chosen to be as large as possible while maintaining enough data points in the point cloud.



Supplementary Figure 2. Visualisation of embedded point clouds via standard TDE and cyclic TDE. The plots show PCA projections of the embedded point clouds in 3D with various embedding dimensions and time delays. In each plot, the percentages along an axis represent the explained variance ratio of a PCA eigenvalue. The colour legends show locations of sampled data points in the time series prior to embedding. Standard TDE yields sparse and scattered points when both embedding dimension d and time delay τ are large. The number of sampled points that get embedded is constrained by these parameters, hence the varying colour legends in (a) and (c). **a** TDE with fixed $d = 10$. Time delay τ varies over $\{5, 10, 50, 100\}$, but cannot increase significantly further. **b** CTDE with fixed $d = 10$. Time delay τ varies over a bigger range including $\{5, 100, 500, 1000\}$, which demonstrates greater flexibility. **c** TDE with fixed $\tau = 1$. Embedding dimension d varies over $\{10, 100, 500, 1270\}$. Notably, when d reaches 1270, the point cloud gaps, without forming a closed cycle (i.e., head-to-tail connection). Consequently, no significant 1-dimensional MP can be captured in the phonetic time series. In fact, when the embedding dimension further increases to 1290, an empty 1-dimensional barcode is obtained due to the lack of points necessary to form even a single cycle. **d** CTDE with fixed $\tau = 1$. Embedding dimension d again varies over $\{10, 100, 500, 1270\}$, which showcases the method's improved entirety and faithfulness in representing topological characteristics.

With a proper dimension, we then compute the delay parameter for the embedding. According to Perea and Harer [8], in the case of a periodic function, the optimal delays τ can be expressed as

$$\tau = m \cdot \frac{T}{d} \quad (1)$$

where T denotes the (minimal) period, d represents the dimension of the embedding, and m is a positive integer.

Under these conditions, we obtain the theoretically optimal MP. However, the time series under consideration in our experiments are far from periodic. As a result, we use

the first peak of the ACL function to represent the period T . The common choice of τ is to let the window size $\tau \cdot d$ equal the (minimal) period. However, in the case of a discrete time series, one often obtains $\tau = 0$ or $\tau = 1$ in this way, since the embedding dimension d is too large in comparison. Therefore, one strategy is to increase m to get a relatively reasonable τ . Here we set $m = 6$. Then the resulting τ from Supplementary Equation (1) is rounded to the nearest integer (if it equals 0, take 1 instead). The performance of MP with delay obtained in this way is presented in Supplementary Sec. 5.3 below.

It is not uncommon that the window size exceeds the number of points in a time series, resulting in an empty embedding. Let $|S|$ denote this number of (sampled) points, i.e., the data-point capacity of the time series. In the extreme case where $|S| < 500$, we apply our variant of cyclic TDE (see Supplementary Sec. 3.3 and Discussion in the main text), with d and τ chosen as above. Otherwise, if $500 \leq |S| \leq \tau \cdot d$, we adopt the delay $\tau' = |S|/d$ instead, rounded downwards. These measures enable us to obtain an appropriate delay for each time series, facilitating the attainment of significant MP for a specified homological dimension.

Lastly, we let skip equal 5. We choose this value for the skip parameter mainly to reach a satisfactory computation time. The impact of the skip parameter in TDE on MP and computation time is further shown in Supplementary Sec. 5.2 below.

Once we set the parameters, the time series are transformed into point clouds. If the number $|P|$ of points in a point cloud (to be distinguished from the number $|S|$ above of points sampled from the time series prior to TDE) is less than 40, we consider this sample to be invalid, with too few points to represent the original structure of the time series. Note that this necessary threshold applies to both training and test data for machine learning (cf. Fig. 3e).

4.3 Computing and vectorising persistent homology

Using Ripser [12, 13] for PH (see Supplementary Sec. 2 and Introduction in the main text), we compute the PDs of the point clouds in a fast and efficient way. We then extract MP from each 1-dimensional PD, using persistence birth time and lifetime as two features of a time series.

The process of vectorising a PD presents a challenge due to the indeterminate (and potentially large) number of intervals in the barcode, coupled with the ambiguous information they contain. This ambiguity arises from our lack of knowledge about the types of information that can be derived from different parts of the PD.

Here, we only extract the MP and corresponding birth time. This decision is informed by our prior selection of an appropriate set of parameters for TDE, which ensures that the MP reaches its theoretical optimum (see Discussion for discussion and experiments beyond optimising MP theoretically).

5 DISCUSSION ON PARAMETER SELECTION

Here, supplementing Discussion in the main text and Supplementary Sec. 4.2, we provide further details for a more balanced, comprehensive, and justified perspective. In the studies applying topological methods to analyse time series [14, 15, 16, 17, 18, 19, 20], the determination of parameters for TDE emerges as a pivotal aspect. This stems from the significant impact that the selection of parameters has on the resulting topological spaces and their corresponding PDs. There exist several convenient algorithms for parameter selection. For example, the False Nearest Neighbours algorithm, a widely utilised tool, provides a method for deciding the minimal embedding dimension [21]. However, in the context of PH, usually the objective is not to achieve

a minimal dimension. Contrarily, a dimension of substantial magnitude may be desirable due to certain advantages it offers.

5.1 Embedding dimension and maximal persistence

In the TDE-PH approach, the determination of dimension in a TDE can be complex. However, it plays a pivotal role in the extraction of topological descriptors such as MP. It is observed that a larger dimension can significantly enhance the theoretically optimal MP of a time series. In TopCap, the dimension of TDE is set to be 100, a relatively large dimension for the experiment. On the other hand, several factors also constrain this choice. These include the length of the sampled time series, since the dimension cannot exceed the length, otherwise it would render the resulting point cloud literally pointless. The constraints also include the periodicity of the time series, as the time-delay window size should be compatible with the approximate period of the time series, at least theoretically as in Supplementary Equation (1).

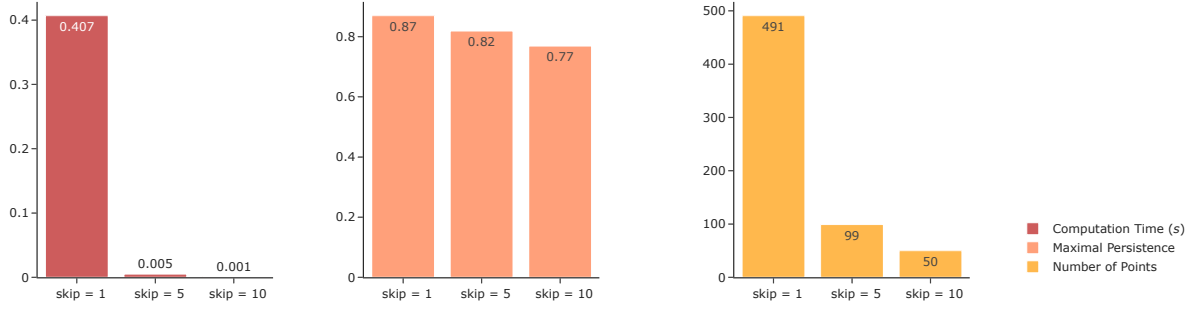
According to Perea and Harer [8, Proposition 5.1], there is no information loss for trigonometric polynomials if and only if the dimension of TDE exceeds twice the maximal frequency. Here, no information loss implies that the original time series can be fully reconstructed from the embedded point cloud. In general, for a periodic function, a higher dimension of TDE can yield a more precise approximation by trigonometric polynomials. Although there are no absolutely periodic functions in real data, each time series exhibits its own pattern of vibration, as discussed in the last part of Results in the main text, and a higher dimension of embedding may be employed to capture a more accurate vibration pattern in the time series.

Moreover, an increased embedding dimension may result in reduced computation time for PD. For instance, computation times for the voiced consonant [ɪ] in Fig. 8a are 0.2671, 0.2473, and 0.2375 seconds, corresponding to embedding dimensions 10, 100, and 1000. This is attributed to the reduction on the number of points in the embedded point cloud due to a higher dimension (cf. CTDE from Discussion in the main text and Supplementary Sec. 3.3). While this reduction in computation time may not be considered substantial compared to the impact of changing skip (see Supplementary Sec. 5.2 below), it may become significant when handling large datasets.

More importantly, an increased embedding dimension can yield benefits such as enhanced MP (cf. Fig. 8c), which serves as a major motivation for our choice of higher dimensions, as well as a smoother shape of resulting point clouds obtained through TDE, which makes the embedding visibly reasonable.

Typically, for most algorithms, a lower dimension is preferred due to factors such as those associated with the curse of dimensionality and computational cost. By contrast, in TopCap, we opt instead for a higher dimension.

However, the embedding dimension cannot be arbitrarily large. As illustrated in Supplementary Fig. 2c, when the embedding dimension escalates to 1270, it becomes unfeasible to capture a significant MP in the phonetic time series. This results from a break of the point cloud. When



Supplementary Figure 3. **Computation duration, MP, and the size of point clouds as the skip parameter increases.** An increase in skip can lead to a significant reduction in computation time, owing to the reduced size of the point cloud. However, MP remains relatively stable.

dimension = 10 theoretical delay = 40			dimension = 50 theoretical delay = 8			dimension = 100 theoretical delay = 4		
delay	skip	MP	delay	skip	MP	delay	skip	MP
1	1	0.0610	1	1	0.2834	1	1	0.4270
10	1	0.1299	3	1	0.3021	2	1	0.4337
20	1	0.1312	4	1	0.3054	2	5	0.4146
30	1	0.1281	5	1	0.3058	3	1	0.4357
39	1	0.1229	6	1	0.3042	3	5	0.4120
39	5	0.1134	7	1	0.3052	4	1	0.4381
40	1	0.1290	7	5	0.2886	4	5	0.4139
40	5	0.1195	8	1	0.3093	5	1	0.4375
41	1	0.1200	8	5	0.2928	5	5	0.4105
41	5	0.1153	9	1	0.3091	6	1	0.4347
45	1	0.0940	9	5	0.2913	6	5	0.4114
50	1	0.1226	10	1	0.3069	7	1	0.4380
60	1	0.1315	15	1	0.3070	8	1	0.4378
94	1	empty	18	1	empty	9	1	empty

Supplementary Table 1. **1-Dimensional MP for choices of dimension, delay, and skip in TDE.** The theoretical delay is computed as in Supplementary Sec. 4. Empty in MP means that the delay is too large to obtain point-cloud data with proper topological features.

the embedding dimension further reaches 1280, an empty 1-dimensional barcode is obtained due to the lack of points necessary to form even a single cycle. In this way, the dimension of TDE is related to the length of the time series.

5.2 Skip, maximal persistence, and computation time

Computation time assumes a critical role when a computer processes a substantial volume of data. In this context, the parameter skip in TDE significantly influences the number of points within the point clouds, and consequently the number of simplices during persistence filtration and the computation time for PD.

Here, we demonstrate that an appropriate increment in the skip parameter can markedly reduce computation time. Meanwhile, it is noteworthy that MP exhibits stability with respect to an increase in skip. As a result, it is feasible to augment skip in TDE to expedite the computation of PD. On the other hand, for details on the complexity of computing PH, the interested reader may refer to [22, Sec. 4] and [23, Sec. 4.3].

Specifically, using an example with a record of the voiced consonant [m], we elucidate the relationship between skip, computation duration, and size of the resulting point

clouds obtained via TDE in Supplementary Fig. 3. Computation duration is measured each time after restarting the Jupyter notebook, on Dell Precision 3581, with CPU Intel® Core™ i7-13800H of basic frequency 2.50 GHz and 14 cores. Computation time means the time for executing the code `ripser(Points, maxdim=1)`. As depicted in Supplementary Fig. 3, a substantial reduction in computation time is observed with an increase in the skip parameter. In contrast, our computation's output MP appears stable.

5.3 Multiple dependency of maximal persistence

As discussed above, there are three key parameters in TDE, namely, embedding dimension d , delay τ , and skip. In this subsection, we present a table that delineates the topological descriptor MP in relation to these from TopCap.

The experiment is performed on a record of the voiced consonant [ɪ], which comprises 887 sampled points as the length of this time series. Theoretically, given a periodic function, one obtains the optimal MP of the function in a fixed dimension under the condition that the TDE window size (i.e., the product of dimension and delay) equals a period (see Supplementary Sec. 3.1). However, the real-world phonetic time series are far from being periodic. Despite our approach to calculating the period of a time series by

ACL functions, we cannot guarantee that the theoretically derived delay will indeed yield the optimal MP of a time series in general. Nevertheless, this desired delay usually gives relatively good MP.

For instance, as illustrated in Supplementary Table 1, when the dimension is 10, the theoretical delay is 40. This corresponds to an MP of 0.1290, which is marginally lower than the MP of 0.1315 achieved at a delay of 60 (cf. Fig. 8c). However, as the dimension increases, the point clouds from TDE become more regular. It becomes increasingly probable that at the desired delay, one can indeed obtain the optimal MP of the time series. For example, when the dimension is either 50 or 100, the MP of the time series is achieved at the desired delay. This provides additional justification for preference with higher dimensions. Indeed, the table reveals that an augmentation in dimension may lead to a more substantial enhancement in the MP of a time series than simply tuning delay.

REFERENCES

- [1] Gunnar Carlsson et al. "On the local behavior of spaces of natural images". In: *International Journal of Computer Vision* 76 (Jan. 2008), pp. 1–12. DOI: 10.1007/s11263-007-0056-x.
- [2] IPA Chart. *The international phonetic alphabet (revised to 2020)*. International Phonetic Association, retrieved 16th January 2024 from <https://www.internationalphoneticassociation.org/content/ipa-chart>.
- [3] Encyclopædia Britannica. *Model showing the distribution of frequencies along the basilar membrane of the cochlea*. Online. Accessed: 2024-10-09, <https://www.britannica.com/science/inner-ear/media/1/288499/18100>.
- [4] Gurjeet Singh et al. "Topological analysis of population activity in visual cortex". In: *Journal of Vision* 8.8 (June 2008), pp. 11–11. DOI: 10.1167/8.8.11.
- [5] Michael Robinson. *Topological Signal Processing*. Mathematical Engineering. Springer, Heidelberg, 2014, pp. xvi+208. DOI: 10.1007/978-3-642-36104-3.
- [6] Yu Chen, Hongwei Lin, and Jiacong Yan. "The Gestalt computational model by persistent homology". In: *Vision Research* 237.108697 (2025). DOI: <https://doi.org/10.1016/j.visres.2025.108697>.
- [7] Floris Takens. "Detecting strange attractors in turbulence". In: *Dynamical Systems and Turbulence*, Warwick 1980. Ed. by David Rand and Lai-Sang Young. Berlin, Heidelberg: Springer Berlin Heidelberg, 1981, pp. 366–381.
- [8] Jose A. Perea and John Harer. "Sliding windows and persistence: An application of topological methods to signal analysis". In: *Foundations of Computational Mathematics* 15.3 (June 2015), pp. 799–838. DOI: 10.1007/s10208-014-9206-z.
- [9] A. R. Bradlow. (n.d.) ALLSTAR: Archive of L1 and L2 Scripted and Spontaneous Transcripts And Recordings. Retrieved from <https://speechbox.linguistics.northwestern.edu/allstar>.
- [10] Michael McAuliffe et al. "Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi". In: *Proc. Interspeech 2017*. 2017, pp. 498–502. DOI: 10.21437/Interspeech.2017-1386.
- [11] Paul Boersma and David Weenink. *Praat: Doing phonetics by computer*. Version 6.3.09, retrieved 2nd March 2023 from <http://www.praat.org/>. 2023.
- [12] Christopher Tralie, Nathaniel Saul, and Rann Bar-On. "Ripser.py: A Lean Persistent Homology Library for Python". In: *The Journal of Open Source Software* 3.29 (Sept. 2018), p. 925. DOI: 10.21105/joss.00925.
- [13] Ulrich Bauer. "Ripser: Efficient computation of Vietoris-Rips persistence barcodes". In: *Journal of Applied and Computational Topology* 5.3 (2021), pp. 391–423. DOI: 10.1007/s41468-021-00071-5.
- [14] Kenneth Brown and Kevin Knudson. "Nonlinear statistics of human speech data". In: *International Journal of Bifurcation and Chaos* 19 (July 2009), pp. 2307–2319. DOI: 10.1142/S0218127409024086.
- [15] Saba Emrani, Thanos Gentimis, and Hamid Krim. "Persistent homology of delay embeddings and its application to wheeze detection". In: *IEEE Signal Processing Letters* 21.4 (2014), pp. 459–463.
- [16] Cássio MM Pereira and Rodrigo F de Mello. "Persistent homology for time series and spatial data clustering". In: *Expert Systems with Applications* 42.15-16 (2015), pp. 6026–6038.
- [17] Firas A Khasawneh and Elizabeth Munch. "Chatter detection in turning using persistent homology". In: *Mechanical Systems and Signal Processing* 70 (2016), pp. 527–541.
- [18] Lee M Seversky, Shelby Davis, and Matthew Berger. "On time-series topological data analysis: New data and opportunities". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 2016, pp. 59–67.
- [19] Yuhei Umeda. "Time series classification via topological data analysis". In: *Information and Media Technologies* 12 (2017), pp. 228–239.
- [20] Marian Gidea and Yuri Katz. "Topological data analysis of financial time series: Landscapes of crashes". In: *Physica A: Statistical Mechanics and its Applications* 491 (2018), pp. 820–834.
- [21] Holger Kantz and Thomas Schreiber. *Nonlinear Time Series Analysis*. Vol. 7. Cambridge University Press, 2004.
- [22] Edelsbrunner, Letscher, and Zomorodian. "Topological persistence and simplification". In: *Discrete & Computational Geometry* 28.4 (Nov. 2002), pp. 511–533. DOI: 10.1007/s00454-002-2885-2.
- [23] Afra Zomorodian and Gunnar Carlsson. "Computing persistent homology". In: *Discrete & Computational Geometry* 33.2 (Feb. 2005), pp. 249–274. DOI: 10.1007/s00454-004-1146-y.